Safety Case Report

AURIX™ TC39x BC-Step Microcontroller

This is applicable for the following products:

Product Name	Package/Variants
AURIX™ TC39x BC-Step Microcontroller	*Refer to Section 2.1, Table 1*

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1 Executive Summary

The *AURIX™ TC39x BC-Step Microcontroller* was developed in compliance with ISO26262:2011 (ISO26262 in rest of the document) and it fulfills it's top level safety requirements.

Internal assessments and audits confirm that the evidence and argumentation are available demonstrating *AURIX™ TC39x BC-Step Microcontroller* fulfilling the intent of applicable ISO26262 requirements up to ASIL D.

This safety case report provides arguments and evidence that the *AURIX™ TC39x BC-Step Microcontroller* conforms to the top level safety requirements under the provided use-cases as stated in the Safety Manual for the scope stated in Section 2.1.

An overview of the ISO26262 parts which have been tailored out completely or partially due to inapplicability is given in Section 3.

A concentrated description how the Infineon product development process is carried out to ensure that the *AURIX™ TC39x BC-Step Microcontroller* variants (see Section 2.1) conform to the top level safety requirements is described in Section 4.

A visualized overview in Section 5 provides the arguments and evidence that the development and verification work-products conform to the intent of ISO26262 requirements. A declaration of conformance is provided in Section 6.

A list of known open problems / deviations is given in Open Problems.

2 Introduction

2.1 Scope and Purpose

This Safety Case Report applies to the *AURIX™ TC39x BC-Step Microcontroller* product variants given in below table.

Table 1 **AURIX™ TC39x BC-Step Microcontroller**

Component	Variants	
Hardware	SAL-TC399XX-256F300S	BC
	SAL-TC399XP-256F300S	BC
	SAL-TC397XP-256F300S	BC
	SAK-TC399XP-256F300S	BC
	SAK-TC399XX-256F300S	BC
	SAK-TC397XP-256F300S	BC
	SAK-TC397XA-256F300S	BC
	SAK-TC397XT-256F300S	BC
	SAK-TC397QA-160F300S	BC
	SAK-TC397XX-256F300S	BC
	SAK-TC397QP-192F300S	BC
	SAK-TC397QP-256F300S	BC
	SAK-TC397XZ-256F300S	BC
	SAK-TC397TT-256F300S	BC
Software Units	Software Built-in Self-Test (SBST) for CPU and SPU	

This report serves as the work product requested by the ISO26262 Part 2-6.5.3 and follows the guidance of ISO26262, Part 10-5.3. The assumed or agreed top level safety requirements as given to the integrator of the *AURIX™ TC39x BC-Step Microcontroller* in the Safety Manual are documented in Section 4.

2.2 Product Description

The product description is covered by the User Manual. Technical descriptions such as safety mechanisms, diagnostic coverages and failure rates are documented in the Safety Manual and the Safety Analysis Summary Report.

3 ISO26262 Tailoring for AURIX™ TC39x

The table below gives an overview of the parts of ISO26262 which were tailored out due to inapplicability.

Table 2 **ISO26262 Tailoring Overview**

Part of ISO26262	Requirements References	Tailoring	Keyword list of tailoredout parts (for details see Compliance Matrix)
ISO26262 - 2	2-1 to 2-7.5	Tailored partially	Hazard and Risk Analysis (HARA) is done on item level, Proven in Use Argument is not applicable.
ISO26262 - 3	3-1 to 3-8.5	Tailored out completely	Outside of Infineon scope of activities. Not relevant, because the product is not an item.
ISO26262 - 4	4-1 to 4-Annex A	Tailored partially	Requirements applicable to item/system level only; All software relevant requirements; Validation related requirements.
ISO26262 - 5	5-1 to 5-Annex E	Tailored partially	Requirements applicable to system level; Requirements related to production, operation, service and decommissioning (5-7.5.4); Clauses 5-9.4.3 - Evaluation of each cause of safety goal violation (supported only on customer request).
ISO26262 - 6	6-1 to 6-Annex D	No Tailoring, Fully IN	
ISO26262 - 7	7-1 to 7-Annex A	Tailored partially	Preproduction; Operation, service, maintenance & repair; Decommissioning; Safety relevant transport & storage; Item level requirements
ISO26262 - 8	8-1 to 8-Annex B	Tailored partially	Supplier selection, Supplier project plan/safety plan, HARA; item/integrator level; method "Increased confidence from use" not applied at IFX for SW tool evaluation / qualification; Development interface agreement (DIA), Qualification according 8-12 & 8-13 not used. Proven in use argument according 8-14 not used
ISO26262 - 9	9-1 to 9-Annex A	No Tailoring, Fully IN	
ISO26262 – 10	10-1 to 10-Annex B	Informative	

4 Top Level Safety Requirements and Process Overview

In this section the top level safety requirements are listed as stated in the Safety Manual.
A basic description is given how the product development process work-products support their achievement.

The ISO26262 Compliance Matrix describes in detail how the ISO26262 requirements are fulfilled by arguments and evidences as parts of the product development process).

4.1 Safety Related Product Development Steps: Overview

Infineon's internal product development process (i.e. SDHB) conforms to the objective of the safety measures and methods as described in ISO26262 V-model in overall product development and fulfilling of the top level safety requirements.

Product requirements describe the needed functionality of the microcontroller where as a Safety Concept (based on the assumed Application Use-Cases) provides the necessary technical safety requirements (i.e. requirements for the safety mechanisms) to protect the safety-related functions.

A Fault Tree Analysis (FTA) gives a deductive description of the interdependencies and the implementation of safety mechanisms. A Dependent Failure Analysis (DFA) analyzes the possible occurence of cascading and common cause faults and documents the respective solutions.

Safety-related functions are protected by safety mechanisms against random hardware faults. The Failure Modes, Effects and Diagnostics Analysis (FMEDA) analyzes the effect of possible random hard error and soft error faults on the safety requirements. The diagnostic coverages of safety mechanisms which are in place to protect important functional blocks are focused on achieving the assumed or agreed hardware architectural metrics (failure rate λ for each top level safety requirement), which are outcome of the FMEDA, as specified by the Safety Concept. The achieved diagnostic coverages are documented in the FMEDA.

It shall be noted that the fault detection capabilities of the safety mechanisms and the fault reactions of the *AURIX™ TC39x BC-Step Microcontroller* are fully dependent on the correct configuration by the integrator. This particularly applies to the fault detection and reaction time. Each safety mechanism has a fault detection time but the timing of the complete fault detection and reaction path is determined by the configuration of the safety mechanisms and the ECU reactions by the integrator.

Infineon provides guidance by assumptions of use (see Safety Manual). The ASIL level of these assumptions of use and their implementation is dependent on the specific application and is in the responsibility of the integrator.

Concerning the adequate avoidance of systematic faults, all functions, i.e. nominal functions and safety mechanisms, have been developed according to the Infineon product development flow SDHB which covers the respective ISO26262 requirements.

All respective verification measures are part of the SDHB requirements and the results are documented adequately.

Requirements management as well as configuration and change management are implemented as part of the general Infineon process requirements.

4.2 Top Level Safety Requirements for the AURIX™ TC39x BC-Step Microcontroller Product

This section shows the top level safety requirements (TLSRs) which are applicable for the AURIX™ TC39x BC-Step Microcontroller product. These TLSRs along with the use-cases as elabotrated in the Safety Manual define the safety claims for the AURIX™ TC39x BC-Step Microcontroller product.

The product is developed as a Safety Element out of Context. Hence the final AURIX™ TC39x BC-Step Microcontroller product FIT rate for a customer specific application needs to be evaluated by the integrator. The FMEDA provides customers with ability to perform this evaluation.

The terms shown in [xxx] in the top level safety requirements are explained in the table of glossary below:

Table 3 Glossary of terms used in the Top Level Safety Requirements

Term	Meaning

Term	Meaning
[MCU]	Microcontroller unit
[MCU-TLSR]	A MCU Top-level Safety Requirement (MCU-TLSR) is a Black-box Product-level [MCU] functional safety requirement that specifies functional safety properties of the [MCU] according to: Application Assumptions.
[detect]	In the [MCU-TLSR] the term [detect] is defined as a reference to the [Failure] detection and [Failure] indication activities.
[hardware elements]	In the [MCU-TLSR] a Hardware Element is defined as a ISO26262-1 1.32 [element] definition applied to hardware only.
[software element]	In the [MCU-TLSR] a Software Element is defined as a ISO26262-1 1.32 [element] definition applied to software only.
[MCU Function]	In the [MCU-TLSR], the MCU Function is defined as a [MCU] Functional Scope.
[Software Execution Platform]	In the [MCU-TLSR], the Software Execution Platform is defined as the [MCU] functional scope related to [Processing Unit], [Non-Volatile Memory], [Volatile Memory]. NOTE: The exact scopes for [Processing Unit], [Non-Volatile Memory], [Volatile Memory] can be found in the Safety Manual.
[Internal MCU Communication]	In the [MCU-TLSR], the Internal MCU Communication is defined as the [MCU] functional scope related to the control and data flows of software executed by [Processing Unit] with [hardware elements] of the [MCU] involved in the processing of [MCU Input Interfaces] and [MCU Output Interfaces]. NOTE: The exact scopes for [Processing Unit], [MCU Input Interfaces] and [MCU Output Interfaces] can be found in the Safety Manual.
[Digital Actuation]	In the [MCU-TLSR], the Digital Actuation is defined as the [MCU] functional scope related to [Timer Outputs], [GPIO Interface]. NOTE: The exact scopes for [Timer Outputs] and [GPIO Interface] can be found in the Safety Manual.
[Analog Acquisition]	In the [MCU-TLSR], the Analog Acquisition is defined as the [MCU] functional scope related to [Analog Inputs]. NOTE: The exact scope for [Analog Inputs] can be found in the Safety Manual.
[Digital Acquisition]	In the [MCU-TLSR], the Digital Acquisition is defined as the [MCU] functional scope related to [Timer Inputs], [GPIO Interface]. NOTE: The exact scope for [Timer Inputs] and [GPIO Interface] can be found in the Safety Manual.
[Sensor Acquisition]	In the [MCU-TLSR], the Sensor Acquisition is defined as the [MCU] functional scope related to [SENT Interface], [PSI5 Interface].

Term	Meaning
[Inbound External Communication]	In the [MCU-TLSR], the Inbound External Interface is defined as the [MCU] functional scope related to [Flexray Interfaces], [Ethernet Interfaces], [CAN Interface], [HSSL Interface], [SPI Interface].
[Outbound External Communication]	In the [MCU-TLSR], the Outbound External Interface is defined as the [MCU] functional scope related to [Flexray Interfaces], [Ethernet Interfaces], [CAN Interface], [HSSL Interface], [SPI Interface].
[External Failure Reporting Interface]	In the [MCU-TLSR], the External Failure Reporting Interface is defined as the ability of the [MCU] to enable safety mechanisms to communicate with external hardware devices (e.g. TLF35884) for the purpose of indicating the presence of a [Failure]. The External Failure Reporting Interface indicates two states: the [MCU_FAULT_STATE] or the [MCU_FAULT_FREE_STATE].
[ADAS Processing]	In the [MCU-TLSR], the ADAS Processing shall be defined as the [MCU] functional scope related to [PRQ Context Diagram] [Radar Interface].
[MCU_FAULT_FREE_STATE]	In the [MCU-TLSR], the MCU_FAULT_FREE_STATE is defined as the state of the [MCU] when no [Failure] has been detected by a safety mechanism AND the [MCU] operates according to the functional specification AND according to the intended [application software] configuration.
[MCU_FAULT_STATE]	In the [MCU-TLSR], the MCU_FAULT_STATE is defined as the state of the [MCU] when a [Failure] has been detected by a safety mechanism.
[Internal Failure Reporting Interface]	In the [MCU-TLSR], the Internal Failure Reporting Interface is defined as the ability of the [HSI] to enable safety mechanisms to communicate with software for the purpose of indicating the presence of a [Failure]. The Internal Failure Reporting Interface indicates two states: the [MCU_FAULT_STATE] or the [MCU_FAULT_FREE_STATE].
[HSI]	Hardware Software Interface
[Alternate External Failure Reporting Interface]	In the [MCU-TLSR], the Alternate External Failure Reporting Interface is defined as the ability of the [MCU] to enable safety mechanisms to communicate with external hardware devices (e.g. TLF35884) for the purpose of indicating the presence of a [Failure]. The Alternate External Failure Reporting Interface indicates two states: either the [MCU_FAULT_STATE] or the [MCU_FAULT_FREE_STATE].
[Diagnostic Test Interval]	In the [MCU-TLSR], the defintion of [Diagnostic Test Interval] is according to the defintion in ISO26262-1 1.25

4.2.1 [MCU-TLSR] Software Execution Platform [ASIL D]

The [MCU] shall [detect] [Failures] of the [hardware elements] involved in the [Software Execution Platform] [MCU Function].

4.2.2 [MCU-TLSR] Software Execution Platform [ASIL B]

The [MCU] shall [detect] [Failures] of the [hardware elements] involved in the [Software Execution Platform] [MCU Function].

4.2.3 [MCU-TLSR] Internal MCU Communication [ASIL D]

The [MCU] shall [detect] [Failures] of the [hardware elements] involved in the [Internal MCU Communication] [MCU Function].

4.2.4 [MCU-TLSR] Digital Actuation [ASIL D]

The [MCU] shall [detect] [Failures] of the [hardware elements] involved in the [Digital Actuation] [MCU Function].

4.2.5 [MCU-TLSR] Digital Actuation [ASIL B]

The [MCU] shall [detect] [Failures] of the [hardware elements] involved in the [Digital Actuation] [MCU Function].

4.2.6 [MCU-TLSR] Analog Acquisition [ASIL D]

The [MCU] shall [detect] [Failures] of the [hardware elements] involved in the [Analog Acquisition] [MCU Function].

4.2.7 [MCU-TLSR] Analog Acquisition [ASIL B]

The [MCU] shall [detect] [Failures] of the [hardware elements] involved in the [Analog Acquisition] [MCU Function].

4.2.8 [MCU-TLSR] Digital Acquisition [ASIL D]

The [MCU] shall [detect] [Failures] of the [hardware elements] involved in the [Digital Acquisition] [MCU Function].

4.2.9 [MCU-TLSR] Sensor Acquisition [ASIL D]

The [MCU] shall [detect] [Failures] of the [hardware elements] involved in the [Sensor Acquisition] [MCU Function].

4.2.10 [MCU-TLSR] Sensor Acquisition [ASIL B]

The [MCU] shall [detect] [Failures] of the [hardware elements] involved in the [Sensor Acquisition] [MCU Function].

4.2.11 [MCU-TLSR] Inbound External Communication [ASIL D]

The [MCU] shall [detect] [Failures] of the [hardware elements] involved in the [Inbound External Communication] [MCU Function].

4.2.12 [MCU-TLSR] Outbound External Communication [ASIL D]

The [MCU] shall [detect] [Failures] of the [hardware elements] involved in the [Outbound External Communication] [MCU Function].

4.2.13 [MCU-TLSR] Coexistence of Software-elements [ASIL D]

The [MCU] shall be able to execute safety-related [software elements] with ASIL A, ASIL B, ASIL C, ASIL D and non-safety-related [software elements] (QM).

4.2.14 [MCU-TLSR] Coexistence of Hardware-elements [ASIL D]

The [MCU] shall implement safety-related [hardware elements] with ASIL A, ASIL B, ASIL C, ASIL D and non-safety-related [hardware elements] (QM).

4.2.15 [MCU-TLSR] Avoidance or Detection of Common Cause Failure [ASIL D]

The [MCU] shall avoid or detect common cause failures between [hardware elements] that could lead to the violation of other [MCU-TLSR]s.

4.2.16 [MCU-TLSR] External Failure Reporting [ASIL D]

The [MCU] shall provide the ability to indicate any [Failure] detected by a hardware-based safety mechanism or by a software-based safety mechanism to the [External Failure Reporting Interface].

4.2.17 [MCU-TLSR] Alternate External Failure Reporting [ASIL D]

The [MCU] shall provide the ability to indicate any failure of POWER SUPPLY or CLOCK that are identified as dependent failures to the [Alternate External Failure Reporting Interface].

4.2.18 [MCU-TLSR] Internal Failure Reporting [ASIL D]

The [MCU] shall provide the ability to indicate any [Failure] detected by a hardware-based safety mechanism or by a software-based safety mechanism to the [application software].

4.2.19 [MCU-TLSR] External Failure Reporting Time [ASIL D]

As soon as a [MCU] hardware safety mechanism detects a [Failure] at time T, the [External Failure Reporting Interface] shall transition from the [MCU_FAULT_FREE_STATE] to the [MCU_FAULT_STATE] latest at time T + 125 microseconds.

4.2.20 [MCU-TLSR] Internal Failure Reporting Time [ASIL D]

As soon as a [MCU] hardware safety mechanism detects a [Failure] at time T, the [Internal Failure Reporting Interface] shall transition from the [MCU_FAULT_FREE_STATE] to the [MCU_FAULT_STATE] latest at time T + 125 microseconds.

4.2.21 [MCU-TLSR] External Failure Reporting Interface State Transition Control [ASIL D]

The [MCU] shall provide the [application software] with the ability to change the state of the [External Failure Reporting Interface] from the [MCU_FAULT_STATE] to the [MCU_FAULT_FREE_STATE] AND from the [MCU_FAULT_FREE_STATE] to the [MCU_FAULT_STATE].

4.2.22 [MCU-TLSR] External Failure Reporting Interface Transition Avoidance [ASIL D]

The [MCU] shall avoid that a random hardware fault or a systematic fault of hardware or a systematic fault of software changes the state of the [External Failure Reporting Interface] from the [MCU_FAULT_STATE] to the [MCU_FAULT_FREE_STATE].

4.2.23 [MCU-TLSR] External Failure Reporting Interface Control by External Signals [ASIL D]

The [MCU] shall change the state of the [External Failure Reporting Interface] to [MCU_FAULT_STATE] if requested by the [Emergency Stop Requests] AND if configured by the [application software].

4.2.24 [MCU-TLSR] ADAS Processing [ASIL C]

The [MCU] shall [detect] [Failures] of the [hardware elements] involved in the [ADAS Processing] [MCU Function].

5 Safety Argumentation on ISO26262

The purpose of this section is to provide the general overview of the arguments and evidences that the *AURIX™ TC39x BC-Step Microcontroller* development and verification work products conform to

*the intent of ISO26262 requirements for the ASIL D development and

*internal Infineon processes as defined by SDHB and

*as derived from technical safety requirements specified in the Safety Concept.

5.1 Notation

The safety case approach follows the guidance of ISO26262-10, clause 5.3. The hierarchical trees that are displayed in Section 5.3 follow the principles of the Goal Structuring Notation (GSN).

The GSN is a visualization comparable with a fault tree. The principle of the GSN is to show all branches of the GSN tree in a subsequently visualized logic.

5.2 User Guide

The basic hierarchy is : "Goal (G)" -> "Strategy Linking Goals (S)" in one or more further hierarchy levels -> "Solution" (Sn = evidence to support the truth of argument).

"C" = context

The logic behind the tree is a consequent walk through the whole path down from the goals (G) to the evidences (Sn). The top level claim G1 is met if the strategy linking goals are fulfilled, and so on, until all branches are closed with solutions.

5.3 GSN Structure for product *AURIX™ TC39x BC-Step Microcontroller*

In the various figures below, the above described GSN structure for *AURIX™ TC39x BC-Step Microcontroller* product version is shown.

Throughout this chapter whenever an Infineon Work Product is referenced, it shall be understood as the work product itself and the corresponding review report (checklist).

Top Level Goal G1 – *MCU as a Safety Element Out Of Context (SEooC) is developed adequately free of unreasonable risk.*

The main goal (G1) is to show that the *AURIX™ TC39x BC-Step Microcontroller* is developed adequately free of unreasonable risk in the context of the product description as described in the User Manual and that as an SEooC it satisfies all the safety goals as per the top level safety requirements as stated in section 4.2.

To achieve the main goal G1, four strategies are defined as shown in Figure 1:

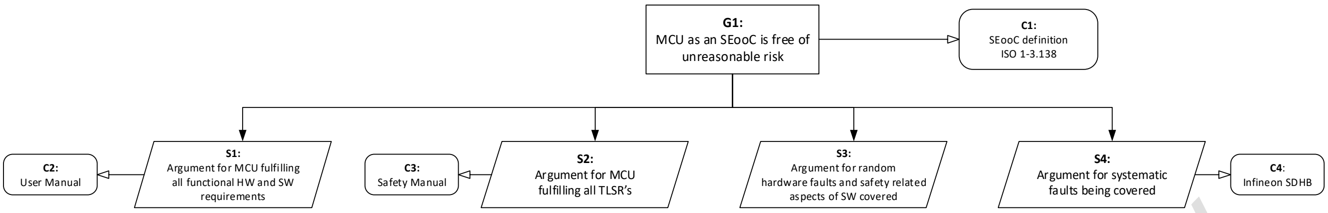

Figure 1: GSN Structure for top level goal G1

For each strategy Sx listed below, a GSN shows the needed detailed goals and functional solutions (Sn) to be implemented into the product.

Table 4 Strategy overview to cover top level goal G1

Strategy	Description
S1	The product has to fulfill all the functional hardware and software requirements and function as per the specifications in the User Manual [8].
S2	The product has to fulfill all the TLSRs as specified in the Product Requirements Document.
S3	Random Hardware Faults (RHF) and safety related aspects of software are covered.
S4	Systematic faults are covered by developing according to the Infineon SDHB process (including process aspects from ISO26262).

5.3.1 Strategy S1 – MCU fulfills all functional HW and SW requirements

In order to fulfil Strategy S1, the below Goals are defined as shown in Figure 2. The justifications by which the defined Goals are claimed to be fulfilled are also shown in the figure.

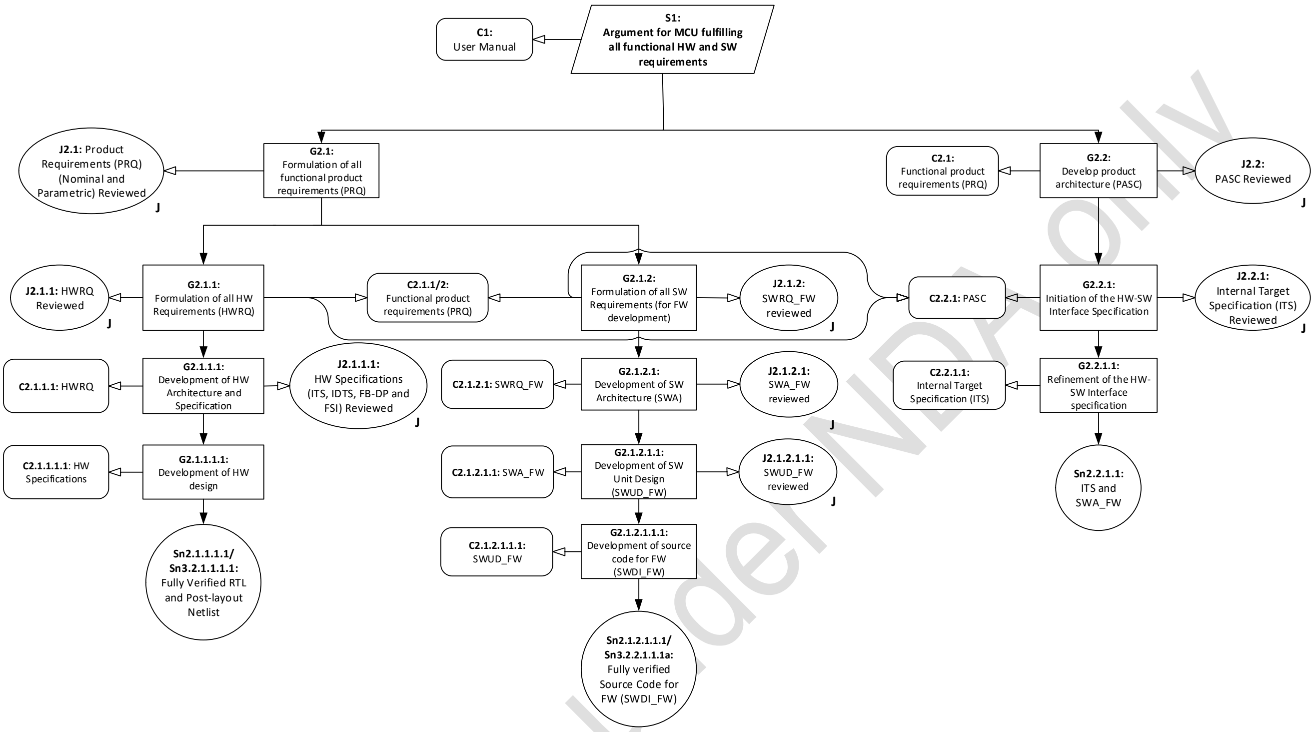

Figure 2: GSN Structure for Strategy S1

5.3.1.1 Goal G2.1 – Formulation of all functional product requirements

The justification that the Goal G2.1 is fulfilled is in the work product - Product Requirements (Nominal & Parametric) [12].

Goal G2.1.1 – *Formulation of all HW Requirements*

The justification that this sub-goal is fulfilled in the context of the Functional Product Requirements (PRQ) and the Product Architecture (PASC) [9] is available in the work product – Hardware Requirements (HWRQ) [13].

Goal G2.1.1.1 – *Development of HW Architecture and Specification*

The justification that this sub-goal is fulfilled in the context of the HW Requirements (HWRQ) is available in the work products – Internal Target Specifications (ITS) [6], Internal Design Target Specifications (IDTS) [7], Functional Block/Design Part mapping (FBDP) [16] and FSI [15].

Goal G2.1.1.1.1 – *Development of HW Design*

The evidence that this sub-goal is fulfilled in the context of the HW Specifications is available in the Solution Sn2.1.1.1.1/Sn3.2.1.1.1.1 - Fully verified RTL and Post-Layout Netlist design.

The Hardware is developed based on the HW specifications and in a subsequent phase the pre-Silicon and post-Silicon verification of the developed Hardware is performed against these specification documents and the corresponding requirements. The verification activities ensure that the implemented hardware fully represents the hardware specification and requirements.

Goal G2.1.2 – *Formulation of all SW Requirements (for FW development)*

The justification that this sub-goal is fulfilled in the context of the Functional Product Requirements (PRQ) and the Product Architecture (PASC) [9] is available in the work product – Software requirements for Firmware (SWRQ_FW) [18].

Goal G2.1.2.1 – *Development of SW Architecture (SWA)*

The justification that this sub-goal is fulfilled in the context of functional requirements for the Firmware is available in the work product - Software Architecture Specification for FW (SWA_FW) [19]. The firmware is developed based on this specification and in a subsequent phase the verification of the developed Firmware is performed against this specification document and against the corresponding requirements.

Goal G2.1.2.1.1 – *Development of SW Unit Design*

The justification that this sub-goal is fulfilled in the context of the SW Architecture is available in the work product – Software Unit Design for Firmware (SWUD_FW) [20].

Goal G2.1.2.1.1.1 – *Development of Source Code for FW*

The evidence that this sub-goal is fulfilled in the context of the SW unit Design for FW is available in the Solution Sn2.1.2.1.1.1/Sn3.2.2.1.1.1a – Fully verified Source Code for Firmware (SWDI_FW) [21].

5.3.1.2 Goal G2.2 – Develop Product Architecture (PASC)

The justification that the goal G2.2 is fulfilled in the context of Functional Product Requirements (PRQ) is available in the work product – Product Architecture and Safety Concept (PASC) [9].

Goal G2.2.1 – *Initiation of the HW-SW Interface Specification*

The justification that this sub-goal is fulfilled in the context of the Product Architecture (PASC) [9] is available in the work product – Internal Target Specification (ITS) [6].

Goal G2.2.1.1 – *Refinement of the HW-SW Interface Specification*

The evidence that this sub-goal is fulfilled is available in the Solution Sn2.2.1.1 – Internal Target Specification (ITS) [6] and Software Architecture Specification for FW (SWA_FW) [19].

5.3.2 Strategy S2 – MCU fulfills all TLSR's

In order to fulfil Strategy S2, the below Goals and sub-goals are defined as shown in Figure 3. The Solutions to fulfil these goals and sub-goals are also shown.

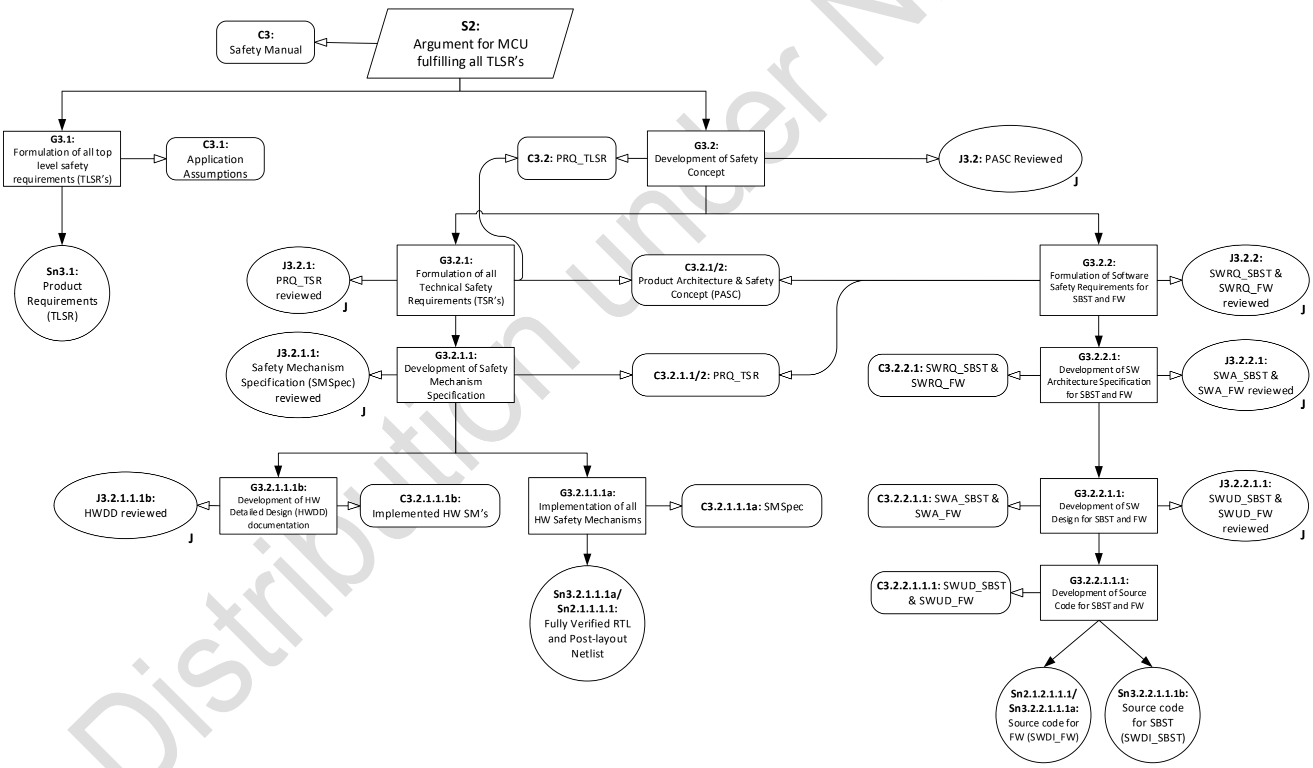

Figure 3: GSN Structure for Strategy S2

5.3.2.1 Goal G3.1 – Formulation of all TLSRs

Since the MCU is a SEooC, the TLSRs are developed in the context of Application Assumptions [5]. The evidence that the Goal G3.1 is fulfilled in the context of the

Application Assumptions is available in the Solution Sn3.1 – Product Requirements (TLSR) [12].

5.3.2.2 **Goal G3.2 – Development of Safety Concept**

The justification that the Goal G3.2, which demonstrates how the TLSRs can be fulfilled on the system level by hardware and software, is fulfilled in the context of the Top Level Safety Requirements (TLSRs). This is available in the work product – Product Architecture and Safety Concept (PASC) [9]. In the Safety Concept, corresponding to each TLSR, certain use-cases are defined which forms the basis of the Safety development for this product. These use-cases are described in the Safety Manual [10] to inform the integrator of all the assumptions under which the safety claims for this product are being made.

Goal G3.2.1 – *Formulation of all Technical Safety Requirements (TSRs)*

The justification that this sub-goal is fulfilled in the context of the TLSRs and the Safety Concept (PASC) [9] is available in the work product - Product Requirements (TSR) [12].

The correctness and completeness of the TSRs to fulfil the TLSRs is verified by means of Inductive and Deductive Safety Analysis. The evidence that such analysis activities were performed are available in the work products – Concept FMEA (CFMEA) [33] and Concept FTA (CFTA) [34].

Goal G3.2.1.1 – *Development of Safety Mechanism Specification*

The justification that the Goal G3.2.1.1 is fulfilled in the context of the TSRs is available in the work product - Safety Mechanism Specification (SMSpec) for the hardware and SBST. The SMSpec is a hybrid workproduct which has both specifications and also Safety requirements for the Hardware and the SBST.

Goal G3.2.1.1.1a – *Implementation of all HW Safety Mechanisms*

The justification that the Goal G3.2.1.1.1 is fulfilled in the context of the SMSpec is available in the Solution Sn3.2.1.1.1a/Sn2.1.1.1.1 - Fully verified RTL and Post-layout Netlist. The correctness and effectiveness of the implemented HW Safety Mechanisms is verified by appropriate safety verification methods as stated in the Safety Plan and PVPL [22] e.g. directed and statistical fault injection methods.

Goal G3.2.1.1.1b – *Development of Hardware Detailed Design*

The justification that the Goal G3.2.1.1.1b is fulfilled in the context of the implemented HW Safety Mechanisms in the MCU is available in the work product – Hardware Detailed Design (HWDD) [17]. This work product is the documentation of the Safety Mechanism implementation inside the Hardware.

Goal G3.2.2 – *Formulation of SW Safety Requirements for SBST and FW*

The TSRs at the product level are refined to formulate the detailed Software Safety Requirements necessary for the FW and SBST development Development. The evidence that this sub-goal is fulfilled in the context of the TSRs at the product level and the Safety Concept is available in the Solution Sn3.2.3, work product – SWRQ_SBST [18] and SWRQ_FW [18].

Goal G3.2.2.1 – *Development of Software Architecture Specification*

The justification that the Goal G3.2.2.1 is fulfilled in the context of the SW Safety Requirements for SBST and FW is available in the work product – Software Architecture Specification for FW (SWA_FW) and Software Architecture Specification for SBST (SWA_SBST).

Goal G3.2.2.1.1 – *Development of SW Design for SBST and FW*

The justification that the Goal G3.2.2.1.1 is fulfilled in the context of the SW Architecture for SBST and FW is available in the work product – Software Unit Design for Firmware (SWUD_FW) and Software Unit Design for SBST (SWUD_SBST).

Goal G3.2.2.1.1.1 – *Development of Source Code for SBST and FW to implement the Software Design*

The evidence that the Goal G3.2.2.1.1.1 is fulfilled is available in the Solution Sn2.1.2.1.1.1/Sn3.2.2.1.1.1a - Source Code for Firmware (SWDI_FW) and Solution Sn3.2.2.1.1.1b – Source Code for SBST (SWDI_SBST)

5.3.3 Strategy S3 – RHF and safety related aspects of SW are covered

In order to fulfil Strategy S3:

- how the MCU is able to detect/control Random Hardware Faults in order to meet HW architectural metrics
- develop a Software free from Systematic faults,

the below Goals and sub-goals are defined as shown in Figure 4. The Solutions to fulfil these goals and sub-goals are also shown.

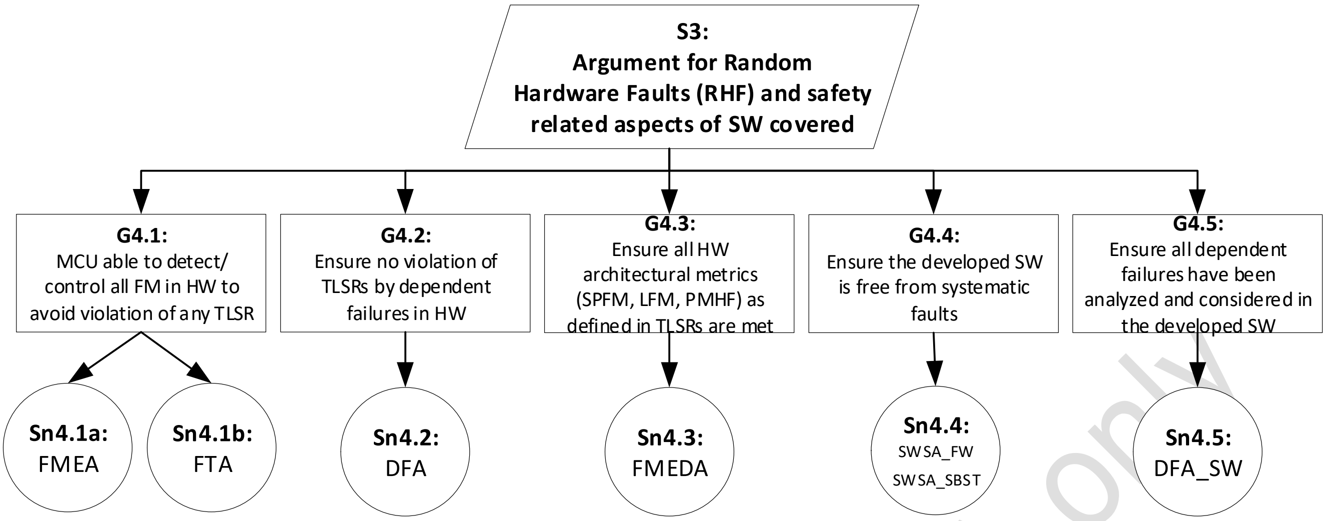

Figure 4: GSN Structure for Strategy S3

5.3.3.1 Goal G4.1 – MCU is able to detect/control all failure modes in the HW to avoid violation of any TLSR

In order to fulfil this goal, exhaustive qualitative safety analyses (both deductive and inductive) were performed to check if all identified failure modes are sufficiently covered by Safety Mechanisms. The evidence that this Goal 4.1 is fulfilled is available in the Solution Sn4.1a – FMEA [35] and Solution Sn4.1b – FTA [36]. In addition to the analysis of random hardware faults, the FMEA work product also covers the analysis of systematic faults in the development process and documents the measures necessary to avoid such systematic faults (in DFMEAs)

5.3.3.2 Goal G4.2 – Ensure no violation of TLSRs and TSRs by dependent faults in HW

In order to fulfil this goal, exhaustive Dependent Failure Analysis (DFA) was performed on the developed hardware to identify all the dependent failure initiators that could bypass or invalidate a required independence or freedom from interference between given hardware elements and thereby violate a safety requirement or a safety goal. Following this investigation, evidence is provided in the Solution Sn4.2 – DFA [37] that all the identified dependent failure initiators are sufficiently covered by safety mechanisms and/or dedicated measures.

5.3.3.3 Goal G4.3 – Ensure all the HW architectural metrics (SPFM, LFM, PMHF) as defined in the TLSRs are met

The evidence that this goal G4.3 is fulfilled is available in the Solution Sn4.3 – FMEDA [40]. The FMEDA documents the HW architectural metrics that were achieved for each TLSR and thereby demonstrates that the metric requirements are met.

5.3.3.4 Goal G4.4 – Ensure the developed SW is free from systematic faults

The evidence that this goal G4.4 is fulfilled is available in the Solution Sn4.4 – Software Safety Analysis for SBST (SWSA_SBST) [38] and for FW (SWSA_FW) [38]. This work product contains detailed analysis of the SBST and FW functions for systematic faults and failures that could lead to the violation of the TLSRs and then identifies the corresponding safety measures to avoid the identified systematic failures during the SW development lifecycle.

5.3.3.5 Goal G4.5 – Ensure all dependent failures have been analyzed and considered in the developed SW

The evidence that this goal G4.5 is fulfilled is available in the Solution Sn4.5 – DFA-SW [39].

The results of all performed safety analyses (CFMEA, CFTA, FMEA, FMEDA, SWSA) is documented in the SAR [41] and summarized further in SASR [42] for the purpose of sharing with the integrator of the MCU HW and SW.

5.3.4 Strategy S4 – MCU is developed free from Systematic Faults

In order to fulfil Strategy S4, the below Goals and sub-goals are defined as shown in Figure 5. The Solutions to fulfil these goals and sub-goals are also shown.

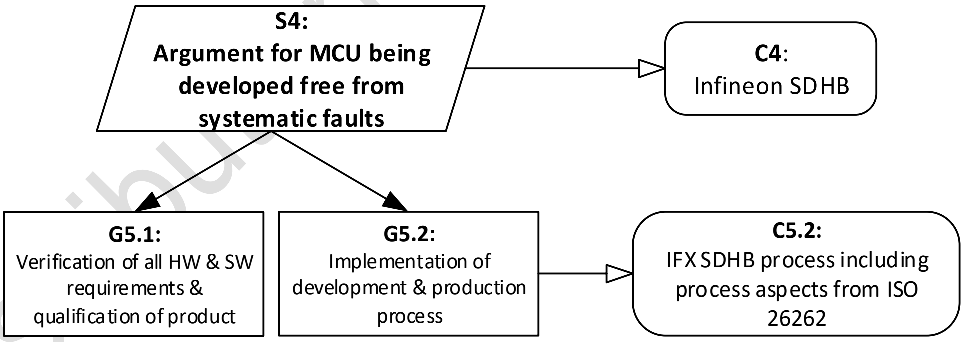

Figure 5: GSN Structure for Strategy S4

5.3.4.1 Goal G5.1 – Verification of all HW and SW requirements and qualification of product

The Goal G5.1 is broken down into further strategies and goals as shown in the Figure 6:

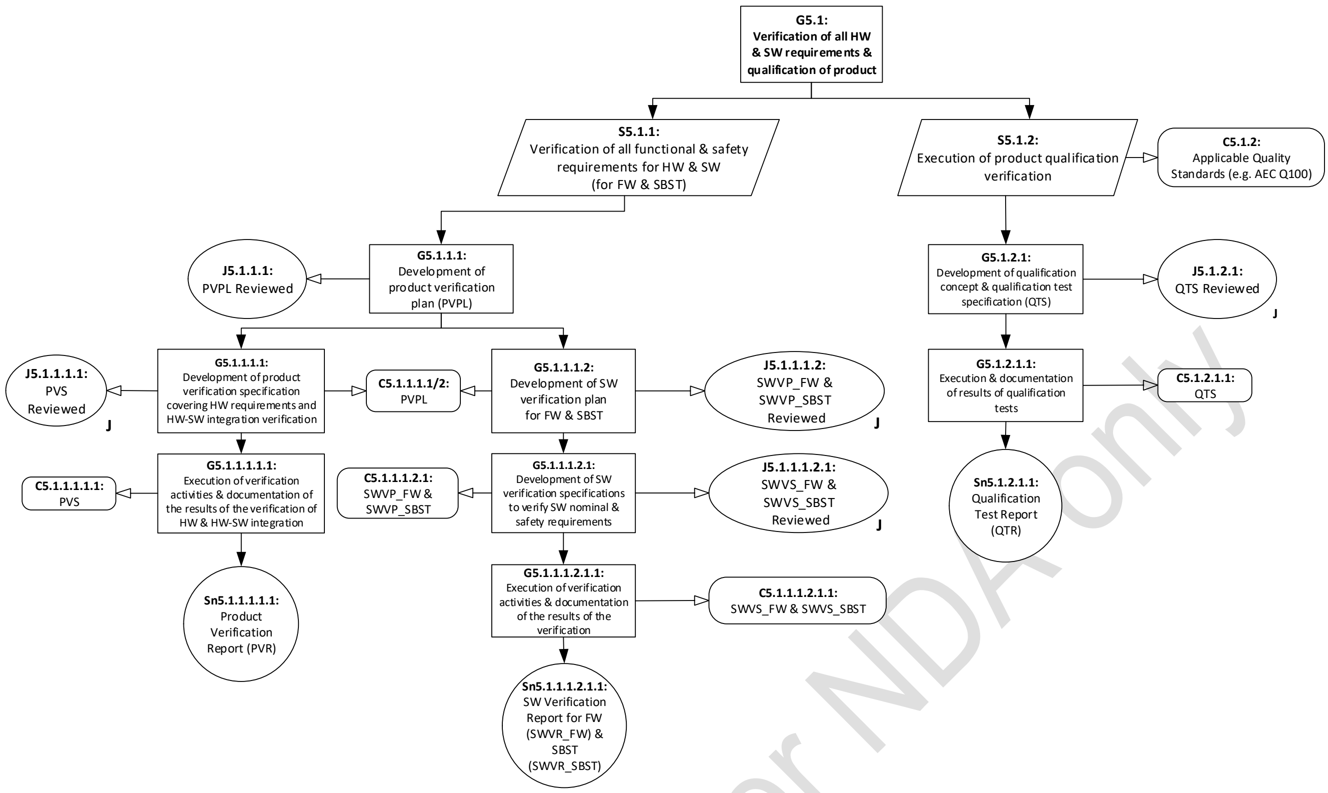

Figure 6: GSN Structure for Strategy S4, Goal G5.1

5.3.4.1.1 Strategy S5.1.1 – Verification of all functional and safety requirements for HW and SW (for FW and SBST)

In order to realize the Strategy S5.1.1, the below Goals are defined:

Goal G5.1.1.1 – *Development of Product Verification PLan (PVPL)*
The PVPL describes the verification strategy, the verification methods and various verification domains needed to correctly verify the nominal and safety functionality of the hardware and software including steps necessary to correctly verify the integration of the hardware and software. The justification that this goal is fulfilled is available in the work product– Product Verification Plan (PVPL) [22].
The Verification Assignment work product allocates the correct and necessary verification domains to all requirements in the PRQ [12], HWRQ [13] and SMSpec [14] to ensure that the nominal and safety requirements are correctly verified.

Goal G5.1.1.1.1 – *Development of Product Verification Specifications covering HW requirements and HW-SW integration verification*
The purpose of this work product is to correctly specify all the testcases that would be needed to fully verify all the nominal and safety hardware requirements and verify the integration of the SW components (FW and SBST) with the hardware. The PVS [25] describes how the requirements

and the content of the HW Specifications in the ITS [6], IDTS [7] and SMSpec [14] are verified and includes both pre-silicon and post-silicon verification domains. The justification that this sub-goal is fulfilled in the context of the PVPL [22] is available in the work product – Product Verification Specification (PVS) [25].

Goal G5.1.1.1.1.1 – *Execution of verification activities and documentation of the results of the verification of the HW and HW-SW integration*
The evidence that this sub-goal is fulfilled in the context of the PVS [25] is available in the Solution Sn5.1.1.1.1.1 – Product Verification Report (PVR) [27]. The PVR summarizes the results of the test cases specified in the PVS [25]. The PVR has an unambiguous statement of pass /fail result and reasons for non-execution of any verification steps and exclusions.

Goal G5.1.1.1.2 – *Development of SW Verification Plan for FW and SBST*
The justification that this sub-goal is fulfilled in the context of the PVPL [22] is available in the work product – Software Verification Plan for FW (SWVP_FW) [24] and for SBST (SWVP_SBST) [24]. These documents contains description of the test methods used at different phases of FW and SBST development in order to sufficiently verify the software requirements.

Goal G5.1.1.1.2.1 – *Development of SW Verification Specifications to verify the SW nominal and safety requirements*
The justification that this sub-goal is fulfilled in the context of the SWVP_FW [24] and SWVP_SBST [24] is available in the work product – Software Verification Specification for FW (SWVS_FW) [26] and Software Verification Specification for SBST (SWVS_SBST) [26].

Goal G5.1.1.1.2.1.1 – *Execution of verification activities and documentation of the results of the verification of the developed Software (for FW and SBST)*
The evidence that the Goal G5.1.1.1.2.1.1 is fulfilled in the context of the SWVS_FW [26] and SWVS_SBST [26] is available in the Solution Sn5.1.1.2.1.1 – Software Verification Report for FW (SWVR_FW) [28] and for SBST (SWVR_SBST) [28]. The SWVR contains detailed results and observed behavior for all the executed software test cases as specified in the SWVS.

5.3.4.1.2 Strategy S5.1.2 – Execution of Product Qualification verification

In order to implement the strategy S5.1.2, the below Goals are defined:

Goal G5.1.2.1 – *Development of Qualification Concept and Qualification Test Specification*
The scope of the Reliability Qualification Concept is to take all necessary actions into account to ensure that the product can be expected to give a sufficient level of reliability in the intended applications and can claim qualification according AEC Q100 (Rev. H) for packaged integrated circuits.
Additionally this Qualification concept also includes the qualification of specific product requirements beyond AEC Q100 Rev H and AEC-Q006. The Qualification Test Specification contains the specification of the tests carried out for qualification, according to AEC Q100 (Rev. H). The justification that Goal G5.1.2.1 is fulfilled is available in the work product – Qualification Test Specification (QTS) [31].

> **Goal G5.1.2.1.1 –** *Execution and documentation of the results of qualification tests*
> The evidence that Goal G5.1.2.1.1 is fulfilled in the context of QTS is available in the Solution Sn5.1.2.1.1 – Qualification Test Report (QTR) [32].

5.3.4.2 Goal G5.2 - Implementation of development and production process

The Goal G5.2, according to Infineon SDHB process, including ISO26262 aspects, is broken down into further strategies and goals as shown in the Figure 7:

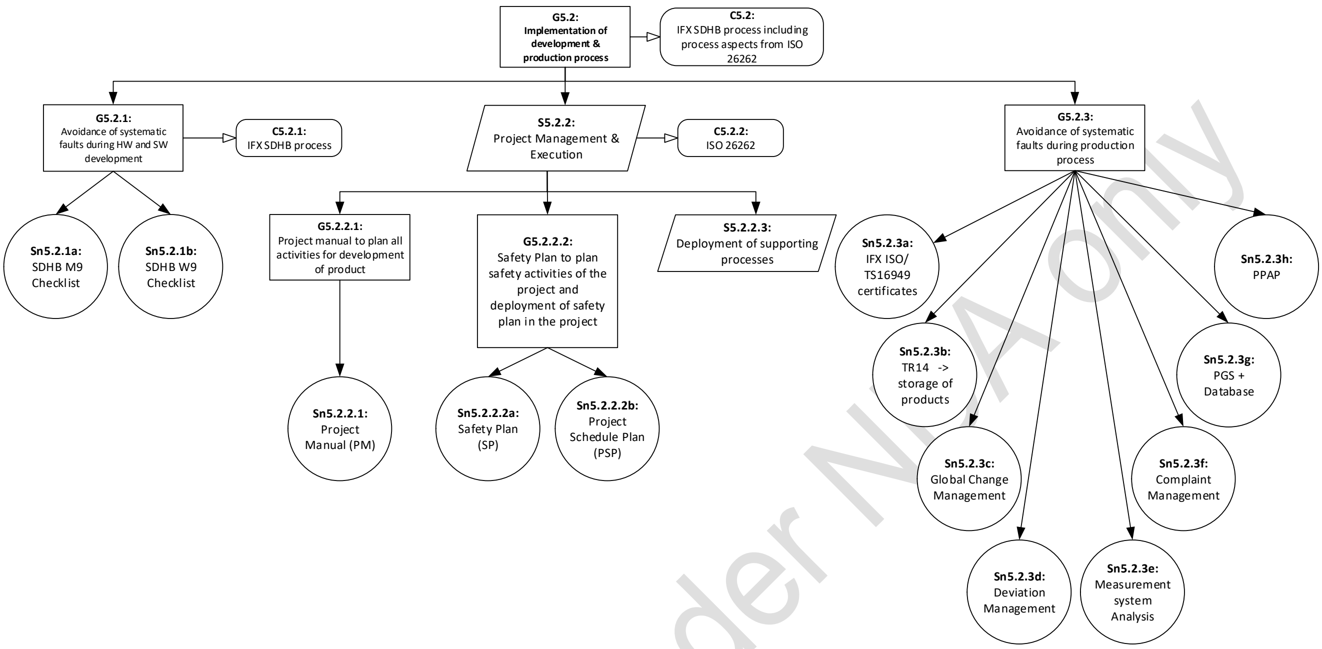

Figure 7: GSN Structure for Strategy S4, Goal G5.2

5.3.4.2.1 Goal G5.2.1 – Avoidance of systematic faults during HW and SW development

The description of the development process is available in the work product - Development Process Documentation (DPD) [43]. The DPD is an umbrella document that provides information related to the following topics:

- Development Process for HW and SW (Infineon SDHB)
- Development Environment for HW and SW
- Requirement Management (RQMP) [11]

The evidence that the Goal G5.2.1 is fulfilled is available in the Solution Sn5.2.1a – SDHB M9 Checklist [55] & Sn5.2.1b – SDHB W9 Checklist [56].

5.3.4.2.2 Strategy S5.2.2 – Project Management and Execution

In order to realize the strategy S5.2.2, the below goals are defined:

Goal G5.2.2.1 – *Project Manual to plan all the activities necessary for the development of the product*

The Project Manual documents all the project management activities which are needed to be planned to start the development cycle for the project. The project organization including the independence needed for conducting audits and assessments for safety is described.
The evidence that this goal is fulfilled is available in the Solution Sn5.2.2.1 – Project Manual (PM) [1].

Goal G5.2.2.2 - *Safety Plan (according to ISO26262) to plan the safety activities of the project and deployment of the Safety Plan in the project*
The evidence that this goal is fulfilled is available in the Solution Sn5.2.2.2a – Safety Plan [2]. The Safety Plan describes the tailoring of the ISO26262 standard and provides argumentations for tailored out Work Products from the standard. The Safety Plan defines the Infineon work products necessary to be developed and addresses the specific safety activities needed for compliance with the tailored ISO26262 standard for the defined scope of the project (in this case HW, FW and SBST). The Safety Plan describes the project dependent safety related aspects and refines project independent safety related aspects already described in the Project Manual (PM) whenever needed.

The evidence that the development of the Work products according to the Safety Plan were planned and executed is available in the Solution Sn5.2.2.2b – Project Schedule Plan (PSP) [3]. The PSP defines all activities with respect to delivery dates, milestones, tasks, deliverables, responsibilities and resources required for the creation, review and release of functional safety work products. The PSP is maintained throughout the project and is used for monitoring of the progress of the safety activities.

Strategy S5.2.2.3 – *Deployment of supporting processes*
In order to realize this strategy the below goals are defined as shown in Figure 8

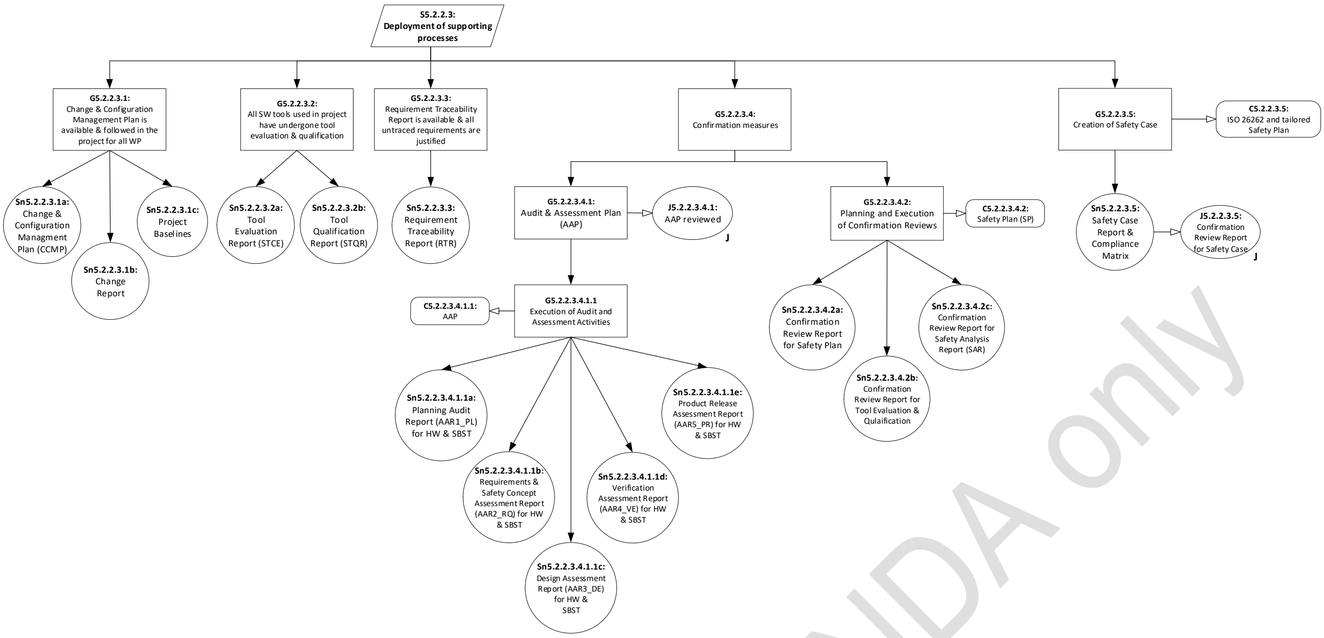

Figure 8: GSN Structure for Strategy S4, sub-strategy S5.2.2.3

Goal G5.2.2.3.1 – *Change and Configuration Management is available and is followed in the project for all work products*
The evidence that this goal is fulfilled is available in the Solutions Sn5.2.2.3.1a – Change and Configuration management Plan (CCMP) [4], Sn5.2.2.3.1b – Change Report & Sn5.2.2.3.1c – Project Baselines.

Goal G5.2.2.3.2 – *All Software tools used in the project have undergone a tool evaluation and qualification*
The evidence that this goal is fulfilled is available in the below Solutions:

- Sn5.2.2.3.2a – Tool Evaluation Report (STCE) [51]
- Sn5.2.2.3.2b – Tool Qualification Report (STQR) [52]

Goal G5.2.2.3.3 – *Requirement Traceability Report is available and all untraced requirements are justified*
The evidence that this goal is fulfilled is available in the Solution Sn5.2.2.3.3 – Requirement Traceability Report (RTR) [44]. This work product provides quantitative evidence about requirements relationships to requirements verification status (horizontal traceability) for all requirements and the evidence about relationships for higher-level requirements from lower-level requirements including the allocation to its architectural elements (vertical traceability)

Goal G5.2.2.3.4 – *Confirmation measures*
This Goal is further broken down into the below Goals:

Goal G5.2.2.3.4.1 – *Creation of Audit and Assessment Plan to plan the Audit and Assessment activities in the project*
The justification that this Goal is fulfilled is available in the work product – Audit and Assessment Plan (AAP) [45].

Goal G5.2.2.3.4.1.1 – *Execution of Audit and Assessment Activities*
The evidence that this Goal is fulfilled in the context of the AAP is available in the below Solutions:

- Sn5.2.2.3.4.1.1a – Planning Audit Report (AAR1_PL) [46] for HW and SBST
- Sn5.2.2.3.4.1.1b – Requirements and Safety Concept Assessment Report (AAR2_RQ) for HW and SBST [47]
- Sn5.2.2.3.4.1.1c – Design Assessment Report (AAR3_DE) for HW and SBST [48]
- Sn5.2.2.3.4.1.1d – Verification Assessment Report (AAR4_VE) for HW and SBST [49]
- Sn5.2.2.3.4.1.1e – Production Release Assessment Report (AAR5_PR) for HW and SBST (combined) [50]

Goal G5.2.2.3.4.2 – *Planning and Execution of Confirmation Reviews*
The evidence that this Goal is fulfilled in the context of the Safety Plan is available in the below Solutions:

- Sn5.2.2.3.4.2a – Confirmation Review Report for Safety Plan (confirmation review report as part of overall work product) [2]
- Sn5.2.2.3.4.2b – Confirmation Review Report for Tool Evaluation and Qualification (confirmation review report as part of overall work product) [51]/[52]
- Sn5.2.2.3.4.2c – Confirmation Review Report for Safety Analysis (confirmation review report as part of overall work product) [41]

Goal G5.2.2.3.5 – *Creation of the Safety Case to confirm the ISO26262 compliance of the product according to the tailoring in Safety Plan*
The evidence that this goal is fulfilled is available in the Solution Sn5.2.2.3.5 – Safety Case Report and Compliance Matrix.

The Compliance Matrix which is part of the Safety Case provides evidences on how the requirements from ISO26262 according to the Safety Plan are fulfilled by the relevant Infineon Work Products.

5.3.4.2.3 Goal G5.2.3 – Avoidance of systematic faults during Production Process

The evidence that this Goal is fulfilled is available in the below Solutions:

- Sn5.2.3.a – IFX ISO/TS16949 certificates
- Sn5.2.3.b – TR14 -> storage of products

- Sn5.2.3.c – Global Change Management
- Sn5.2.3.d – Deviation Management
- Sn5.2.3.e – Measurement system Analysis
- Sn5.2.3.f – Complaint Management
- Sn5.2.3.g – PGS + Database
- Sn5.2.3.h – PPAP

5.4 Audit and Assessments

Table 5 Performed Audit and Assessments

Type	Scope	Start Date	End Date	Status	Rating
Planning Audit (ISO26262)	*AURIX™ TC39x BC-Step Microcontroller*	3/7/2017	6/7/2017	Conducted	Conditional Pass
	SBST (CPU)	24/5/2017	2/5/2018	Conducted	Conditional Pass
	SBST (SPU)	24/5/2017	9/10/2018	Conducted	Conditional Pass
Requirements and Safety Concept Assessment (ISO26262)	*AURIX™ TC39x BC-Step Microcontroller*	27/11/2018	28/11/2018	Conducted	Conditional Pass
	SBST (CPU)	10/10/2018	12/10/2018	Conducted	Conditional Pass
	SBST (SPU)	10/10/2018	12/10/2018	Conducted	Conditional Pass
Design Assessment (ISO26262)	*AURIX™ TC39x BC-Step Microcontroller*	04/01/2019	07/01/2019	Conducted	Conditional Pass
	SBST (CPU)	15/10/2018	16/10/2018	Conducted	Conditional Pass
	SBST (SPU)	15/10/2018	16/10/2018	Conducted	Conditional Pass
Verification Assessment (ISO26262)	*AURIX™ TC39x BC-Step Microcontroller*	08/01/2019	09/01/2019	Conducted	Conditional Pass
	SBST (CPU)	17/10/2018	18/10/2018	Conducted	Conditional Pass
	SBST (SPU)	17/10/2018	18/10/2018	Conducted	Conditional Pass
Production Release Assessment (ISO26262)	*AURIX™ TC39x BC-Step Microcontroller & SBST*	10/01/2019	16/01/2019	Conducted	Pass

6 Declaration of Conformance

Table 6 **List of the conformant ISO26262 Parts**

Part of ISO26262	Requirements References	Degree of Fulfillment
ISO26262 – 2	2-1 to 2-7.5	Complete fulfillment of the relevant requirements (according to the applied tailoring; see Section 3)
ISO26262 – 3	3-1 to 3-8.5	Not relevant, because the product is not an item.
ISO26262 – 4	4-1 to 4-Annex A	Complete fulfillment of the relevant requirements (according to the applied tailoring; see Section 3)
ISO26262 – 5	5-1 to 5-Annex E	Complete fulfillment of the relevant requirements (according to the applied tailoring; see Section 3)
ISO26262 – 6	6-1 to 6-Annex D	Complete fulfillment of the relevant requirements (according to the applied tailoring; see Section 3)
ISO26262 – 7	7-1 to 7-Annex A	Complete fulfillment of the relevant requirements (according to the applied tailoring; see Section 3)
ISO26262 – 8	8-1 to 8-Annex B	Complete fulfillment of the relevant requirements (according to the applied tailoring; see Section 3)
ISO26262 – 9	9-1 to 9-Annex A	Complete fulfillment of the relevant requirements (according to the applied tailoring; see Section 3)

7 References

The following documents were used as applicable input references:

Table 7 Applicable Reference Documents

No.	Title	Issue Date	Version
[1]	ISO 26262:2011, Road Vehicles — Functional Safety	15.11.2011	1st Edition
[2]	GSN Community Standard	November 2011	Version 1

Below table lists the internal documents referenced in this safety case report as evidence. The internal documents listed below are baselined according to the following internal baseline: **TC39x_A2GProduct_M9_v1**

Table 8 Reference List of Work Products and Internal Documents

Ref.	Work Product	Abbr.	Release Manager Container
[1]	Project Manual	PM	MC_PRJDOC_PM
[2]	Safety Plan[1]	SP	MC_PRJDOC_SP
[3]	Project Schedule Plan	PSP	MC_PRJDOC_PSP_FS
[4]	Change and Configuration Management Plan	CCMP	MC_PRJDOC_CCM
[5]	Application Assumptions	AA	MC_ACE_PRJDOC_AA
[6]	Internal Target Specification	ITS	MC_ACE_PRJDOC_ITS
[7]	Internal Design Target Specification	IDTS	MC_PRJDOC_IDTS
[8]	User Manual (including Data Sheet and Errata Sheet)[2]	UM	MC_ACE_PRJDOC_UM MC_ACE_PRJDOC_DS MC_ACE_PRJDOC_ES MC_PRJDOC_SWDI_SBST_CPU MC_PRJDOC_SWDI_SBST_SPU
[9]	Product Architecture and Safety Concept[1]	PASC	MC_ACE_PRJDOC_PASC
[10]	Safety Manual[2]	SM	MC_ACE_PRJDOC_SM
[11]	Requirements Management Plan	RQMP	MC_ACE_PRJDOC_RQMP
[12]	Product Requirements	PRQ	MC_ACE_PRJDOC_PRQ
[13]	Hardware Requirements	HWRQ	MC_ACE_PRJDOC_HWRQ
[14]	Safety Mechanism Specification	SMSpec	MC_ACE_PRJDOC_SMS

[1] Inspection possible at Infineon sites

[2] Provided to the Customer

Ref.	Work Product	Abbr.	Release Manager Container
[15]	Functional and Software Interface	FSI	MC_ACE_PRJDOC_FSI
[16]	Functional Block/Design Part mapping	FBDP	MC_PRJDOC_FBDP
[17]	Hardware Detailed Design	HWDD	MC_PRJDOC_HWDD
[18]	SW Requirements for Firmware SW Requirements for SBST_CPU SW Requirements for SBST_SPU	SWRQ	MC_PRJDOC_SWRQ_FW MC_PRJDOC_SWRQ_SBST_CPU MC_PRJDOC_SWRQ_SBST_SPU
[19]	SW Architecture for Firmware SW Architecture for SBST_CPU SW Architecture for SBST_SPU	SWA	MC_PRJDOC_SWA_FW MC_PRJDOC_SWA_SBST_CPU MC_PRJDOC_SWA_SBST_SPU
[20]	SW Unit Design for Firmware SW Unit Design for SBST_CPU SW Unit Design for SBST_SPU	SWUD	MC_PRJDOC_SWUD_FW MC_PRJDOC_SWUD_SBST_CPU MC_PRJDOC_SWUD_SBST_SPU
[21]	SW Design Implementation for Firmware SW Design Implementation for SBST_CPU SW Design Implementation for SBST_SPU	SWDI	MC_PRJDOC_SWDI_FW MC_PRJDOC_SWDI_SBST_CPU MC_PRJDOC_SWDI_SBST_SPU
[22]	Product Verification Plan	PVPL	MC_PRJDOC_PVPL
[23]	Verification Assignment	VA	MC_PRJDOC_VA
[24]	SW Verification Plan for Firmware SW Verification Plan for SBST_CPU SW Verification Plan for SBST_SPU	SWVP	MC_PRJDOC_SWVP_FW MC_PRJDOC_SWVP_SBST_CPU MC_PRJDOC_SWVP_SBST_SPU
[25]	Product Verification Specification	PVS	MC_PRJDOC_PVS
[26]	SW Verification Spec for Firmware SW Verification Spec for SBST_CPU SW Verification Spec for SBST_SPU	SWVS	MC_PRJDOC_SWVS_FW MC_PRJDOC_SWVS_SBST_CPU MC_PRJDOC_SWVS_SBST_SPU
[27]	Product Verification Report	PVR	MC_PRJDOC_PVR
[28]	SW Verification Report for Firmware SW Verification Report for SBST_CPU SW Verification Report for SBST_SPU	SWVR	MC_PRJDOC_SWVR_FW MC_PRJDOC_SWVR_SBST_CPU MC_PRJDOC_SWVR_SBST_SPU
[29]	Configuration Data Implementation	CDI	MC_CDI_FW
[30]	Configuration Sector (CFS) Documentation	CFD	MC_PRJDOC_CFD
[31]	Qualification Test Specification	QTS	MC_PRJDOC_QTS
[32]	Qualification Test Report	QTR	MC_PRJDOC_QTR
[33]	Concept Failure Mode and Effects Analysis	C-FMEA	MC_ACE_PRJDOC_CFMEA
[34]	Concept Fault Tree Analysis	C-FTA	MC_ACE_PRJDOC_CFTA

Ref.	Work Product	Abbr.	Release Manager Container
[35]	Failure Mode and Effects Analysis	FMEA	MC_PRJDOC_FMEA
[36]	Fault Tree Analysis	FTA	MC_PRJDOC_FTA
[37]	Dependent Failure Analysis	DFA	MC_PRJDOC_DFA
[38]	Software Safety Analysis for Firmware Software Safety Analysis for SBST_CPU Software Safety Analysis for SBST_SPU	SWSA	MC_PRJDOC_SWSA_FW MC_PRJDOC_SWSA_SBST_CPU MC_PRJDOC_SWSA_SBST_SPU
[39]	Dependent Failure Analysis – Software	DFA-SW	MC_PRJDOC_DFASW_FW
[40]	Failure Modes Effects and Diagnostic Analysis[2]	FMEDA	MC_PRJDOC_FMEDA
[41]	Safety Analysis Report	SAR	MC_PRJDOC_SAR
[42]	Safety Analysis Summary Report[2]	SASR	MC_PRJDOC_SASR
[43]	Development Process Documentation	DPD	MC_PRJDOC_DPD
[44]	Requirement Traceability Report	RTR	MC_PRJDOC_RTR
[45]	Audit and Assessment Plan	AAP	MC_PRJDOC_AAP
[46]	Planning Audit Report	AAR1_PL	MC_PRJDOC_AAR_PL MC_PRJDOC_AAR_PL_SBST
[47]	Requirement and Safety Concept Assessment Report	AAR2_RQ	MC_PRJDOC_AAR_RQ MC_PRJDOC_AAR_RQ_SBST
[48]	Design Assessment Report	AAR3_DE	MC_PRJDOC_AAR_DE MC_PRJDOC_AAR_DE_SBST
[49]	Verification Assessment Report	AAR4_VE	MC_PRJDOC_AAR_VE MC_PRJDOC_AAR_VE_SBST
[50]	Production Release Assessment Report	AAR5_PR	MC_PRJDOC_AAR_PR
[51]	Tool Evaluation Report	STCE	MC_PRJDOC_STCE
[52]	Tool Qualification Report	STQR	Included in MC_PRJDOC_STCE
[53]	Embedded Coding Rules for Firmware Embedded Coding Rules for SBST_CPU Embedded Coding Rules for SBST_SPU	ECR	MC_PRJDOC_ECR_FW MC_PRJDOC_ECR_SBST_CPU MC_PRJDOC_ECR_SBST_SPU
[54]	Safety Case Report[2] (including Compliance Matrix)	SCR	MC_PRJDOC_SCR
[55]	SDHB M9 Checklist		MC_PRJDOC_SDHB_TC39x
[56]	SDHB W9 Checklist		MC_PRJDOC_SDHB_SBST_CPU MC_PRJDOC_SDHB_SBST_SPU

Table 9 Reference List of Customer Documentation and Internal Documents

Document	Customer Version	Release Date	Internal Document
AURIX™ TC3xx IFX User Manual	V1.0.0	2018-08	MC_ACE_PRJDOC_UM
AURIX™ TC39x Appendix to User's Manual	V1.0.0	2018-08	MC_ACE_PRJDOC_UM
AURIX™ TC3xx ED Target Specification [1]	V2.5.1	2018-04	MC_ACE_PRJDOC_UM
TC39x_DS_v10	V1.0	2018-11	MC_ACE_PRJDOC_DS
TC39x_Data_Sheet_Addendum_V1.1	V1.1	2018-08	MC_ACE_PRJDOC_DS
AURIX TC3xx Safety Manual v1.03	V1.03	2019-01	MC_ACE_PRJDOC_SM
TC39x_BC_Errata_Sheet_v1_0	V1.0	2018-12-14	MC_ACE_PRJDOC_ES
AURIX_TC39x_FMEDA_v1.01	V1.01	2018-12-17	MC_PRJDOC_FMEDA
CPU SBST User Manual	V1.0.2.1.1	2018-12-14	MC_PRJDOC_SWDI_SBST_CPU
SPU SBST User Manual	V1.0.0.1.14	2019-01-10	MC_PRJDOC_SWDI_SBST_SPU
A2G_AURIX Safety Analysis Summary Report	V1.04	2019-01-07	MC_PRJDOC_SASR
TC39xBC_SafetyCaseReport_V1.0	V1.0	2019-01	MC_PRJDOC_SCR
TriCore TC1.6.2 Core Architecture Manual; Core Architecture (Vol. 1)	V1.0	2017-01-11	Z8F54561158
TriCore TC1.6.2 Core Architecture Manual; Instruction Set (Vol. 2)	V1.0	2017-01-11	Z8F54351040

[1] Distribution under NDA, only relevant for tool development not for application development

8 Open Problems

In preparation of the safety case all open problem reports and change requests have been reviewed and assessed for customer relevance. This table shows the customer relevant open problems.

Table 10 Open Problems List

No.	Open Problem Description	Argument for Acceptance

No.	Open Problem Description	Argument for Acceptance
1	**EVRC Frequency and Phase Synchronization to CCU6/GTM Input** As documented in section "EVRC Frequency and Phase Synchronization to CCU6/GTM Input" in TC3xx UM V1.0.0: • Loss of Synchronization Lock event is triggered if switching frequency range is violated. Loss of Synchronization Lock event is indicated via status bits and interrupt. If this functionality is used in a safety related application, the faults leading to loss of synchronization could be detected by reading the status bit or reacting on the interrupt. This ESM is not part of the Safety Manual.	Current FMEDA provides FIT rate for faults of this functionality if configured to be used in safety related application. This is such small that FIT targets are not violated.
2	**Unexpected SMU Safety FF Alarms** Due to a synchronization issue, ALM6[7] and ALM10[21] is sporadically triggered if the PRE1, PRE2 or TFSP_HIGH fields of register FSP are written while the SMU is configured either in Time Switching protocol (FSP.MODE = 10_B) and FSP[0] is toggling with a defined T_{SMU_FFS} period, or in Dual Rail protocol (FSP.MODE = 01_B) and FSP[1:0] are toggling with a defined T_{SMU_FFS} period. In addition, an unexpected ALM10[16] or ALM10[17] is sporadically triggered if field FSP.PRE1 or RTC.RTD is written, and at least one recovery timer is running based on a defined T_{SMU_FS} period (regardless of the FSP.MODE configuration). The alarms can only be cleared with cold or warm Power-On reset. **Workaround:** To avoid unexpected alarms, configuration of PRE1, PRE2 and TFSP_HIGH is only allowed when FSP is in Bi-stable protocol mode (FSP.MODE = 00_B). Mode switching and configuration shall not be done with the same write access to register FSP. • disable Time Switching or Dual Rail protocol by setting FSP in Bi-stable protocol mode (FSP.MODE = 00_B); • wait until Bi-stable protocol mode is active (read back register FSP twice); • write desired value to PRE1, PRE2 and	In general, unintended SMU alarms are rated as safe. An alarm reaction by means of power on reset might not be possible for all applications. The proposed workaround shall be applied by customers until an errata sheet is available to prevent generation of unintended alarms.

No.	Open Problem Description	Argument for Acceptance
	TFSP_HIGH; • then switch FSP.MODE to the desired protocol (optional step) If the mode shall be changed after writing to one of the fields PRE1, PRE2, or TFSP_HIGH while in Bi-Stable protocol mode (FSP.MODE = 00_B): • write desired value to PRE1, PRE2 and TFSP_HIGH; • then switch FSP.MODE to Time Switching or Dual Rail protocol. If field FSP.PRE1 or RTC.RTD shall be written, make sure no recovery timer is running. It is not allowed to write to the RTD field when at least one recovery timer is running (indicated by bits RTS0 and RTS1 in the STS register).	
3	Customer documentation (FMEDA) does not provide evidence for usage of GPT12 in safety related applications in combination with GTM. However, Safety Manual still shows ESM which refer to GPT12. It is still possible for the integrator to use GPT12 upon own assessment.	Current FMEDA provides FIT rate for faults of GPT12 functionality if used in safety related application. Customer needs to configure the FMEDA accordingly to cope with FIT rate increase by means of customer developed safety mechanism.
4	Customer documentation (FMEDA) does not provide evidence for usage of PORTS as single channel in safety related applications. It is still possible for the integrator to use PORTS as single channel upon own assessment.	Current FMEDA provides FIT rate for faults of PORTS functionality if used as single channel in safety related applications. Customer needs to configure the FMEDA accordingly to cope with FIT rate increase by means of customer developed safety mechanism.
5	Safety Case Report 1.0 is baselined to a requirement database which has minor number of known incorrect relations of safety requirements (< 1%).	Defendable, all incorrect relations analyzed and assessed, no safety related issues found
6	ESM[SW]:CPU:SOFTERR_MONITOR proof to achieve a DC claim of 60% is missing, hence this ESM shall not be used in FMEDA	Customer shall not use this ESM in the FMEDA. FMEDA shows ASIL-B metrics for [MCU-TLSR] Software Execution Platform [ASIL B] can still be achieved even with disabled ESM.
7	TC39x Qualification fails in Humidity Stress Test	Comprehensive safety impact analysis provides evidence, that occurrence of observed fail in the field will not compromise the product safety related aspects.

9 Abbreviations

Table 11 Abbreviations

Abbreviation	Explanation
AoU	Assumption of Use
ASIL	Automotive Safety Integrity Level
AEC	Automotive Excellence Council
DFA	Dependent Failure Analysis
DTI	Diagnostic Test Interval
FIT	Failure In Time
FMEDA	Failure Modes Effects and Diagnostics Analysis
FMEA	Failure Mode and Effects Analysis
FTA	Fault Tree Analysis
FW	Firmware
GSN	Goal Structuring Notation
IFX	Infineon Technologies
ISO	International Organization for Standardization
LFM	Latent Fault Metric
MCU	Microcontroller unit (including Hardware and Software components)
PASC	Product Architecture and Safety Concept Document
PMHF	Probabilistic Metric for Random Hardware Failures
RHF	Random HardWare Faults
SDHB	System Development Handbook (Infineon's Product Development Process)
SEooC	Safety Element out of Context
SPFM	Single Point Fault Metric
SW	Software
SWA	SoftWare Architecture Specification
PTV2	Parametric Requirements
SEooC	Safety Element Out Of Context
SBST	Software Built-in Self Test
FW	Firmware
TLSR	Top Level Safety Requirement
TSR	Technical Safety Requirement

10 Document History

Table 12 Document History

Version	Page or Reference	Description of change	Date of change
V1.00	All	First Release	2019-01

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